Lingyu Li, **Ph.D.**

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CV of Lingyu Li

Bio. **Postdoctoral Fellow** with wide-ranging expertise in **bioinformatics and machine learning research** and extensive experiences in **biomedical data analysis and statistical evaluation**, able to work independently as well as collaboratively within a team.

Education

Sep 2012 – Jun 2016	**B.Sc. of Mathematics and Applied Mathematics**, Shandong Normal University, Jinan, China *Advisor* : Prof. Jinjun Fan **GPA** : 3.51/5.0 | **Rank** : 26/204
Sep 2016 – Jun 2019	**M.Sc. of Computational Mathematics**, Shandong Normal University, Jinan, China *Advisor* : *Prof. Ziwen Jiang* **GPA** : 3.98/5.0 | **Rank** : 1/8
Sep 2019 – Jun 2023	**Ph.D. in Biomedical Engineering**, Shandong University, Jinan, China *Advisor* : *Prof. Zhi-Ping Liu* **GPA** : 94.80/100 | **Rank** : 1/46
Dec 2021 – Jul 2023	**Joint Ph.D. in Bioinformatics**, The University of Hong Kong, Hong Kong SAR, China *Advisor* : *Prof. Wai-Ki Ching* Exchange Program Scholarship Recipient

Research Experience

Aug 2023 – present	**Postdoc in Bioinformatics**, The University of Hong Kong, Hong Kong SAR, China *PI* : *Prof. Yuanhua Huang* Research focus : Spatial transcriptomics and multimodal AI for nucleus-resolved tumor-immune interaction.

Key Publications ([1] denotes First Author, * denotes Corresponding Author)

First or Corresponding Author (latest first) :

1. Qiu Y*, Chen X, Zhu Z, Li L*, Nie Q*. A multitask spatial transcriptomics analysis tool using a smoothed attention graph autoencoder. **Nature Communications**, **Revised**, Jun 2026.
(**JCR : Q1, Rank : 5/200 in the category of Multidisciplinary, IF : 15.7**)
2. Li L, Sun L, Li S, Ching W-K*, Liu Z-P*. Cell dynamics on energy landscapes : Comparing attractor detection in Boolean-network and diffusion-based models under *in silico* gene perturbations, **Quantitative Biology**, Jun 2026.
(**JCR : Q3, Rank : 78/125 in the category of Biochemistry, Genetics and Molecular Biology (miscellaneous), IF : 1.4**)
3. Li L, Zhao W, Zhang Q, Ching W-K, Liu Z-P*. CNet-Cox for interpretable network biomarker discovery and survival risk scoring in precise breast cancer prognosis, **npj Digital Medicine**, May 2026.
(**JCR : Q1, Rank : 3/153 n the category of Medicine, IF : 15.1**)
4. Li L, Wang T, Liang Z, Yu H, Ma S, Yu L*, Huang Y*. FineST : Contrastive learning integrates histology and spatial transcriptomics for nuclei-resolved ligand-receptor analysis, **Nature Communications**, Mar 2026.
(**JCR : Q1, Rank : 5/200 in the category of Multidisciplinary, IF : 15.7**)
5. Li L, Ching W-K, Liu Z-P*. Prognostic biomarker discovery via a connected network-constrained Cox proportional hazards model, **Briefings in Bioinformatics**, vol.27, no.5, Mar 2026.
(**JCR : Q1, 25/474 in the category of Computer Science, IF : 7.7**)
6. Li L, Liu Z-P*. Biomarker discovery from gene expression data by connected network-constrained support vector machine, **Expert Systems with Applications**, vol.226, 120179, Sep 2023.
(**JCR : Q1, Rank : 10/344 in the category of Engineering, IF : 8.665**)
7. Li L, Sun L, Chen G, Wong C-W, Ching W-K*, Liu Z-P*. LogBTF : Gene regulatory network inference using Boolean threshold network model from single-cell gene expression data, **Bioinformatics**, vol.39, no.5, btad256, Apr 2023.
(**JCR : Q1, Rank : 6/293 in the category of Mathematics, IF : 6.937**)
8. Li L, Algabri Y A, Liu Z-P*. Identifying diagnostic biomarkers of breast cancer based on gene expression data and ensemble feature selection, **Current Bioinformatics**, vol.18, no.3, pp.232-246, Mar 2023.
(**JCR : Q1, Rank : 38/201 in the category fo Mathematics, IF : 4.850**)
9. Li L, Ching W-K, Liu Z-P*, Robust biomarker screening from gene expression data by stable machine learning-recursive feature elimination methods, **Computational Biology and Chemistry**, vol.100, 107747, Jul 2022.
(**JCR : Q2, Rank : 56/201 in the category of Mathematics, IF : 3.737**)
10. Li L, Liu Z-P*, A connected network-regularized logistic regression model for feature selection, **Applied Intelligence**, vol.52, no.10, pp.11672-11702, Jan 2022.
(**JCR : Q1, Rank : 86/450 in the category of Computer Science, IF : 5.019 in the year of 2022**)

11. Li L, Liu Z-P*. Detecting prognostic biomarkers of breast cancer by regularized Cox proportional hazards models, **Journal of Translational Medicine**, vol.19, pp.1-20, Dec 2021.
(**JCR : Q1, Rank : 40/668 in the category of Medicine, IF : 8.459**)
12. Li L, Liu Z-P*, Discovery of spontaneous preterm birth biomarkers based on machine learning, **Journal of Nanjing University (Natural Sciences Edition)**, vol.57, no.5, pp.767-774, Sep 2021. (**In Chinese**)
13. Li L, Liu Z-P*, Biomarker discovery for predicting spontaneous preterm birth from gene expression data by regularized logistic regression, **Computational and Structural Biotechnology Journal**, vol.18, pp.3434-3446, Nov 2020.
(**JCR : Q1, Rank : 13/152 in the category of Biochemistry, Genetics and Molecular Biology, IF : 7.271**)
14. Li L, Jiang Z*, Yin Z*, Compact finite-difference method for 2D time-fractional convection–diffusion equation of groundwater pollution problems, **Computational and Applied Mathematics**, vol.39, no.3, 142, May 2020.
(**JCR : Q2, Rank : 136/665 in the category of Mathematics, IF : 2.239**)
15. Li L, Jiang Z*, Yin Z, Fourth-order compact finite difference method for solving two-dimensional convection–diffusion equation, **Advances in Difference Equations**, vol.2018, no.1, pp.1-24, Jul 2018.
(**JCR : Q1, Rank : 1/117 in the category of Mathematics, IF : 2.803**)
16. Li L, Yin Z*, Numerical simulation of groundwater pollution problems based on convection diffusion equation, **American Journal of Computational Mathematics**, vol.7, no.3, pp.350–370, Sep 2017.
17. Li L, Fan J*, Uniform convergence of function term series and its application, **Journal of Shandong Normal University (Natural Science Edition)**, vol.31, no.4, pp.12–19, Dec 2016. (**In Chinese**)

Contributing Author :

1. Wang T, Li L, Huang Y*. SparseAEH : Scalable autoregressive expression histology discovery in spatial transcriptomics via sparse Gaussian kernels, **Nature Methods**, BioRxiv; Submitted, Feb 2026.
(**JCR : Q1, Rank : 2/441 in the category of Biochemistry, IF : 32.1 in the Year 2025**)
2. Wang T, Li L, Liu Z-P*. getDNB : identifying dynamic network biomarkers from time-varying gene regulations utilizing graph embedding techniques, **Briefings in Bioinformatics**, vol.27, 2025.
(**JCR : Q1, Rank : 25/474 in the category of Computer Science, IF : 7.7 in the year of 2025**)
3. Keikha F, Li L, Ching W-K, Liu Z-P*. NetWalkRank : Cancer driver gene prioritization in multiplex gene regulatory networks by a random walk approach, **IEEE-ACM Transactions on Computational Biology and Bioinformatics**, vol.22, 2025.
(**JCR : Q1, Rank : 23/665 in the category of Mathematics, IF : 3.4 in the year of 2025**)
4. Algabri Y A, Li L, Liu Z-P*, scGENA : A single-cell gene co-expression network analysis framework for clustering cell types and revealing biological mechanisms, **Bioengineering-Basel**, vol.9, Jun 2022.
(**JCR : Q3, Rank : 90/163 in the category of Chemical Engineering, IF : 5.046 in the year of 2022**)
5. El Bairi K, Haynes H R, Blackley E, et al. (Rank : 132), The tale of TILs in breast cancer : A report from the international immuno-oncology biomarker working group, **npj Breast Cancer**, vol.7, Jun 2021.
(**JCR : Q1, Rank : 10/346 in the category of Medicine, IF : 7.519 in the year of 2021**)

Patents :

1. Li L, Yang J, Wu C, Gao R, Liu Z-P*. A prognostic biomarker identification system, Page bookmark : CN117352048A, Jan 2024.

Presentations

1. Li L, Yu L, Huang Y*, Hist2Pheno enables nuclei-resolved phenotypic prediction from histological images in esophageal squamous cell carcinoma. in The 34th Conference on Intelligent Systems for Molecular Biology (ISMB), **Top Conferences in Bioinformatics**, **Oral Presentation** 20 mins, Washington, DC, United States, 12-16 Jul 2026.
2. Li L, Ching W-K*, Huang Y*, From gene regulatory dynamics to spatial cellular phenotypes : Learning and modeling across scales. in HKU Workshop on Mathematical Biology, **Oral Presentation** 60 mins, Hong Kong, 15 Jun 2026.
3. Li L, Ching W-K, Liu Z-P*, Prognostic biomarker discovery via a connected network-constrained Cox proportional hazards model. GIW XXXIV ISCB-Asia 2025, **Poster Presentation**, Hong Kong, 10-13 Dec 2025.
4. Li L, Huang Y*, Histology foundation model-derived contrastive learning enables high-fidelity nuclei-resolved spatial ligand receptor interaction. in The 14th National Conference on Bioinformatics and Systems Biology, **Oral Presentation**, Nanning, China, 7-10 Oct 2025.
5. Li L, Liu Z-P, Ching W-K, Huang Y*, Unlocking biomedical data with sparse statistical learning and multimodal artificial intelligence. in The Fourth Mathematical and Life Science Conference, **Oral Presentation**, Beijing, China, 8-10 Aug 2025.
6. Li L, Huang Y*, Nucleus-resolved ligand-receptor interaction discovery by fusing spatial RNA-seq and histology images. in Human Cell Atlas (HCA) Asia 2024 Meeting, **Poster Presentation**, Hong Kong Science Park, Hong Kong, Dec 2024.
7. Li L, Huang Y*, Super-resolved ligand-receptor interaction discovery by fusing spatial RNA-seq and histology images. in The 32nd Conference on Intelligent Systems for Molecular Biology (ISMB), **Top Conferences in Bioinformatics**, **Oral Presentation** 20 mins, Montreal, Canada, 12-16 Jul 2024.
8. Li L, Huang Y*, Fine ligand-receptor interaction discovery by fusing spatial transcriptomics and histology stains. in The 1st HKU-CUHK Joint Postdoctoral Biomedical Sciences Symposium, **Oral Presentation**, Hong Kong, Mar 2024.

9. **Li L**, Liu Z-P*, Identifying diagnostic biomarkers of high-grade serous ovarian cancer based on gene expression data and machine learning methods. in *The Sixth CCF Bioinformatics Conference*, **Oral Presentation**, Qingdao, China, Aug 2021.
10. **Li L**, Liu Z-P*, Discovery of spontaneous preterm birth biomarkers based on machine learning. in *CCF Conference on Artificial Intelligence*, **Poster Presentation**, Yantai, China, Jul 2021.

Project records

Nov 2020 - Nov 2025	**Research on Mathematical Models and Algorithms in Breast Cancer Precision Medicine** › NSFC, National Key R&D Program, NO. 2020YFA0712402. › Participant.
Jan 2020 - Dec 2023	**Research on Bioinformatics Methods for Integrating Multi-level Omics Data to Discover Complex Disease Biomarkers** › NSFC, General Projects, NO. 61973190. › Participant (main).
Jul 2021 - Jun 2023	**Analysis and Prediction of Adverse Pregnancy Outcomes using Peripheral Blood Metabolomics Data of Pregnant Women undergoing Assisted Reproduction** › Collaborate with the Reproductive Medicine Research Center of Shandong University. › Participant (main).

Competitions

Dec 2019	**The 16th Huawei Cup China Postgraduate Mathematic Contest in Modeling (GMCM) - Third Prize** › Submitted paper : Rapid path planning of intelligent aircraft under multiple constraints
Dec 2017	**The 14th Huawei Cup GMCM - Third Prize** › Submitted paper : Flight recovery problem
Dec 2016	**The 13th Huawei Cup GMCM - Second Prize** › Submitted paper : Analysis of genetic locus with inherited diseases and traits

Honors & Awards

July 2026	ISMB 2026 Conference Travel Fellowships from International Society for Computational Biology (**ISCB**)
July 2024	ISMB 2024 Conference Travel Fellowships from International Society for Computational Biology (**ISCB**)
Mar 2023	Shandong University 2023 outstanding graduates
Oct 2022	Shandong University 2022 *Ji Defa* Postgraduate Scholarship for PhD
Sep 2021	The Scholarship under Shandong University's Exchange Program (with **University of Hong Kong**)
Jan 2019	Shandong Normal University 2019 outstanding graduates
Oct 2017	Shandong province 2017 excellent bachelor's degree thesis
Oct 2013	National Encouragement Scholarship

Professional services

Society Member	International Society for Computational Biology (ISCB), China Society for Industrial and Applied Mathematics (CSIAM), Chinese Bioinformatics Society (in preparation) - Network Biology Committee, Program Committee (PC) of IEEE International Conference on Bioinformatics and Biomedicine (BIBM).
Peer Reviewer	*Cell* (1), *Bioinformatics* (2), *Genome Research* (1), *Communications Biology* (1), *iScience* (3), *Small Methods* (2), *Scientific Reports* (4), *Computational Biology and Chemistry* (7), *Briefings in Functional Genomics* (2), *Biology* (2), *Computers in Biology and Medicine* (5), *Computer Methods and Programs in Biomedicine* (4), *BMC Cancer* (2), *Neural Networks* (2). *International journal of molecular sciences* (3). *Biophysics Reports* (2).
Conference Reviewer	*ICIC2024 Conference*(**CCF C**) (3), *ICAI2024 Conference* (3), *IEEE BIBM 2026 Conference* (**CCF B**) (7), etc.

Teaching assistant

Sep 2019–Jan 2020	Teaching Assistant, **Complex Functions and Laplace Transform**, Automation, Shandong University.
Feb 2019–Jun 2019	Teaching Assistant, **Probability Theory and Mathematical Statistics**, Shandong Jiaotong University.
Feb 2019–Jun 2019	Teaching Assistant, **Ordinary Differential Equations**, CS 1701 and 1702, Shandong Normal University.
Feb 2018–Jan 2019	Teaching Assistant, **Advanced Algebra I&II**, Statistics 1801 and 1802, Shandong Normal University.
Sep 2018–Jan 2019	Teaching Assistant, **Advanced Mathematics II**, Economics 1701, Shandong Normal University.
Sep 2017–Jan 2019	Teaching Assistant, **Mathematical Analysis I&II&III**, Mathematics 1703 and 1704, Shandong Normal University.

Skills

Programming Skills :	**R** (5-year), Python (4-year), Matlab (10-year), Linux (3-year), LaTeX(7-year), **C**, Spass
Computer Certificates :	Computer 2-level **C language**, Computer 3-level **Network technology**
Other Certificates :	Putonghua proficiency **second-level A**, Teacher qualification certificate **Mathematics**

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